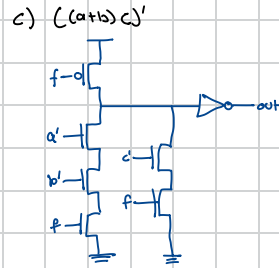
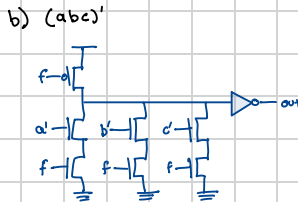
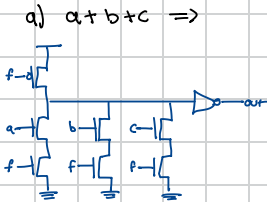


Homework 3

Kaushik
Vada

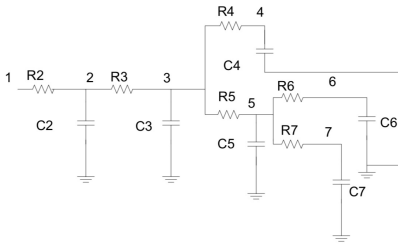
1. (10pt) Draw the transistor-level schematics for domino gates that implement these functions:

- (a) $a+b+c$
- (b) $(abc)'$
- (c) $((a+b)c)'$



2. (10 pts) For the RC circuit shown below, compute the Elmore delays of from

- (a) from node 1 to node 4;
- (b) from node 1 to node 7;
- (c) from node 4 to node 7;



$$a) D_{14} = R_2(C_2 + C_3 + C_4 + C_5 + C_6 + C_7) + R_3(C_3 + C_4 + C_5 + C_6 + C_7) + R_4C_4$$

$$b) D_{17} = R_2(C_2 + C_3 + C_4 + C_5 + C_6 + C_7) + R_3(C_3 + C_4 + C_5 + C_6 + C_7) + R_5(C_5 + C_6 + C_7) + R_7C_7$$

$$c) D_{47} = D_{17} - D_{14} = R_5(C_5 + C_6 + C_7) + R_7C_7 - R_4C_7$$

3. (20pt) Compute the Elmore delay for these wires assuming each wire is divided into 100 sections:

a) Poly wire of width 2λ , length $1,000\lambda$.

b) Metal 1 wire of width 3λ , length $1,000\lambda$.

c) Metal 1 wire of width 3λ , length $10,000\lambda$.

a) (a) Poly Interconnect

Given: $L = 1000\lambda = 90\mu\text{m}, \quad W = 2\lambda = 0.18\mu\text{m}$

Resistance: $R = R_{\text{poly}} \cdot \frac{L}{W} = 8 \cdot \frac{1000\lambda}{2\lambda} = 4000\Omega$

Capacitance: $C = C_{\text{poly,plate}} \cdot (W \cdot L) + C_{\text{poly,fringe}} \cdot 2(W + L)$
 $= 63(90)(0.18) + 63 \times 2(90 + 0.18) = 12383.28\text{ aF}$
 $C = 1.2383 \times 10^{-14}\text{ F}$

Elmore Delay: $\delta_E = \frac{1}{2}rcn(n+1) = \frac{1}{2} \times 40 \times 1.2383 \times 10^{-18} \times 100(101)$
 $\delta_E = 2.50 \times 10^{-11}\text{ s}$

b) (b) Metal 1 Interconnect

Given: $L = 1000\lambda = 90\mu\text{m}, \quad W = 3\lambda = 0.27\mu\text{m}$

Resistance: $R = R_{M1} \cdot \frac{L}{W} = 0.08 \times \frac{1000\lambda}{3\lambda} = \frac{80}{3} \approx 26.7\Omega$

Capacitance: $C = C_{M1,\text{plate}} \cdot A_{M1} + C_{M1,\text{fringe}} \cdot P_{M1}$
 $= 36(90)(0.27) + 54 \times 2(90 + 0.27)$
 $C = 10623.96\text{ aF} = 1.0624 \times 10^{-14}\text{ F}$

Delay: $\delta_E = \frac{1}{2}rcn(n+1)$
 $= \frac{1}{2} \times \frac{8}{30} \times 1.0624 \times 10^{-18} \times 100(101)$
 $\delta_E = 1.43 \times 10^{-13}\text{ s}$

c) (c) Extended Metal 1 Interconnect (Longer Line)

Given: $L = 10000\lambda = 900\mu\text{m}, \quad W = 3\lambda = 0.27\mu\text{m}$

Resistance: $R = R_{M1} \cdot \frac{L}{W} = 0.08 \times \frac{10000\lambda}{3\lambda} = \frac{800}{3} \approx 266.7\Omega$

Capacitance: $C = C_{M1,\text{plate}} \cdot A_{M1} + C_{M1,\text{fringe}} \cdot P_{M1}$
 $= 36(900)(0.27) + 54 \times 2(900 + 0.27)$
 $C = 105977.16\text{ aF} = 1.0598 \times 10^{-13}\text{ F}$

Delay: $\delta_E = \frac{1}{2}rcn(n+1) = \frac{1}{2} \times \frac{8}{3} \times 1.0598 \times 10^{-18} \times 100(101)$
 $\delta_E = 1.43 \times 10^{-13}\text{ s}$

4. (20pt) Compute the optimal number of buffers and buffer sizes for these RC wires when driven by a minimum-size inverter:

a) Poly wire of width 3λ , length $1,000\lambda$.

b) Metal 2 wire of width 3λ , length $10,000\lambda$.

a) (a) Poly Interconnect: $W = 3\lambda, L = 1000\lambda$

First, compute the wire resistance:
 $R_{\text{int}} = R_{\text{poly}} \cdot \frac{L}{W} = 8 \times \frac{1000\lambda}{3\lambda} = \frac{8000}{3} \approx 2666.7\Omega$

Then, the total capacitance per wire segment:
 $C_{\text{int}} = C_{\text{poly,plate}}(WL) + C_{\text{poly,fringe}}[2(W + L)]$
 $C_{\text{int}} = 63(90)(0.27) + 63 \times 2(90 + 0.27) = 12904.92\text{ aF} = 12.9049\text{ fF}$

Now, substitute into the optimization expressions:
 $k_{\text{opt}} = \sqrt{\frac{0.4R_{\text{int}}C_{\text{int}}}{0.7R_0C_0}} = \sqrt{\frac{0.4 \times \frac{8}{3} \times 12.9049}{0.7 \times 6.47 \times 1.78}} = 1.31$
 $h_{\text{opt}} = \sqrt{\frac{R_0C_{\text{int}}}{R_{\text{int}}C_0}} = \sqrt{\frac{6.47 \times 12.9049}{\frac{8}{3} \times 1.78}} = 4.19$

Hence, the optimal number of buffers is approximately $\lceil k_{\text{opt}} = 1.3 \rceil$ and each should be sized about $\lceil h_{\text{opt}} = 4.2 \rceil$ times the minimum inverter.

b) (b) Metal 2 Interconnect: $W = 3\lambda, L = 10,000\lambda$

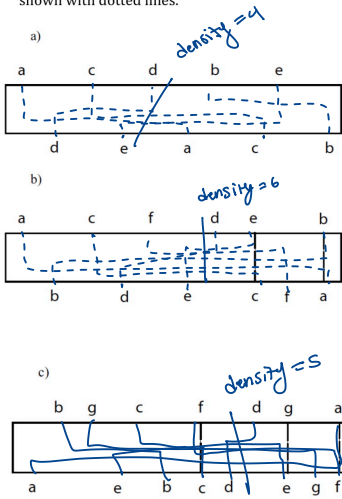
Step 1: Resistance
 $R_{\text{int}} = R_{M2} \cdot \frac{L}{W} = 0.08 \times \frac{10000\lambda}{3\lambda} = \frac{800}{3} \approx 266.7\Omega$

Step 2: Capacitance
 $C_{\text{int}} = C_{M2,\text{plate}}(WL) + C_{M2,\text{fringe}}[2(W + L)]$
 $C_{\text{int}} = 36(900)(0.27) + 51 \times 2(900 + 0.27) = 100575.54\text{ aF} = 100.58\text{ fF}$

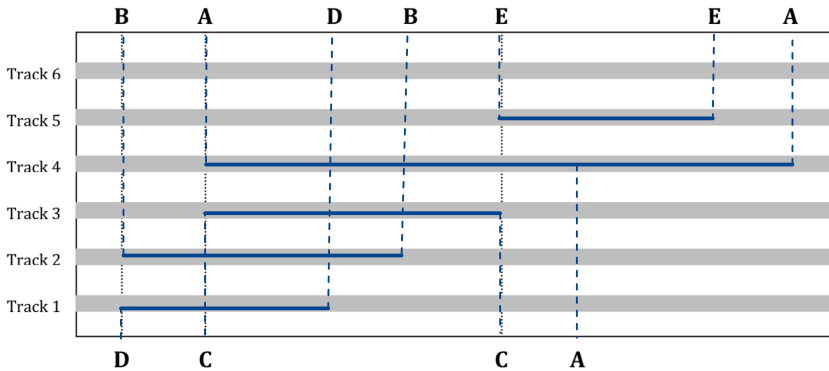
Step 3: Optimal values
 $k_{\text{opt}} = \sqrt{\frac{0.4R_{\text{int}}C_{\text{int}}}{0.7R_0C_0}} = \sqrt{\frac{0.4 \times \frac{800}{3} \times 100.58}{0.7 \times 6.47 \times 1.78}} = 1.15$
 $h_{\text{opt}} = \sqrt{\frac{R_0C_{\text{int}}}{R_{\text{int}}C_0}} = \sqrt{\frac{6.47 \times 100.58}{\frac{800}{3} \times 1.78}} = 37.03$

Result: For the long M2 wire, the ideal buffer count is around $\lceil k_{\text{opt}} = 1.15 \rceil$, with buffer sizes about $\lceil h_{\text{opt}} = 37 \rceil$ times the minimum inverter.

5. (10pt) Compute the density of these channels. Vertically aligned pins are shown with dotted lines.

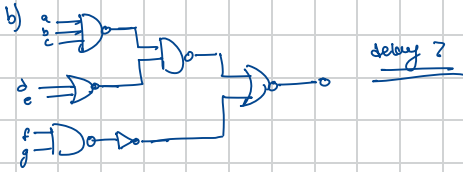
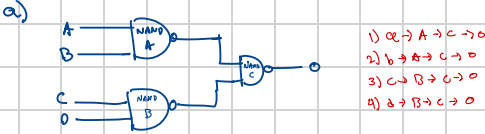


6. (10 pts) For the provided channel (dotted lines show the vertically aligned pins), route the channel using the left-edge algorithm.



7. (10pt) For each of these logic networks, draw the logic diagram and find the critical path, assuming that the delay through all n -input gates are n , delay through an inverter is 1.

- (a) $\text{NAND2}(\text{NAND2}(a,b), \text{NAND2}(c,d))$.
 (b) $\text{AOI21}(\text{NAND3}(a,b,c), \text{NOR2}(d,e), \text{INV}(\text{NAND2}(f,g)))$



8)

8. (10pt) A string of inverters drives a load of 50 pF. Assuming gate capacitance of the input inverter is 18 fF and P_{inv} is 1. To get minimal delay, you are asked to determine the number of inverter stages and sizes following the steps:
- 1) Use $\hat{N} \approx \log_f F$ to estimate the number of stages N (Let $\hat{f} = 4$)
 - 2) Calculate **path delay** and **inverter sizes** (in terms of gate capacitance) using the estimated N .
 - 3) Calculate path delay respectively using $N + 1$ and $N - 1$. Evaluate the estimation accuracy in step 1)

(a) Path effort and stage count estimate

For this path all stages are inverters, so $g_i = 1$ and there is no branching:

$$G = \prod g_i = 1, \quad B = 1.$$

Electrical effort:

$$H = \frac{C_{out}}{C_{in}} = \frac{50 \times 10^{-12}}{18 \times 10^{-15}} = \frac{25000}{9} \approx 2777.78.$$

Thus the path effort is

$$F = GBH = \frac{25000}{9}.$$

Using the heuristic $N \approx \log_f F$ with $f = 4$,

$$\hat{N} \approx \log_4 \left(\frac{25000}{9} \right) = 5.72 \Rightarrow \boxed{N \approx 6}.$$

(b) Delay and stage sizing for $N = 6$

With equal stage effort, $f^* = F^{1/N}$. Take each parasitic $p_i = 1$ (inverters), so $P = \sum p_i = N$.

Path delay.

$$\hat{D} = NF^{1/N} + P = 6 \left(\frac{25000}{9} \right)^{1/6} + 6 = \boxed{28.50} \text{ (in units of } \tau \text{)}.$$

Input capacitances (sizes). Let the first stage be minimum ($C_{in,1} = 18$ fF). Successive stages grow by $F^{1/N}$:

$$C_{in,k} = 18 \text{ fF} \cdot (F^{1/6})^{k-1}, \quad k = 1, \dots, 6.$$

Numerically (reporting the next five since the first is 18 fF):

$$\{ 18F^{1/6}, 18F^{2/6}, 18F^{3/6}, 18F^{4/6}, 18F^{5/6} \} = \{ 67.49, 253.03, 948.68, 3556.89, 13335.84 \} \text{ fF}.$$

(c) Sanity check with $N \pm 1$

Compute the delay using the same equal-effort assumption for neighboring stage counts.

$$D_{N=7} = 7 \left(\frac{25000}{9} \right)^{1/7} + 7 = \boxed{28.73}.$$

$$D_{N=5} = 5 \left(\frac{25000}{9} \right)^{1/5} + 5 = \boxed{29.42}.$$

Since $D_{N=6} = 28.50 < D_{N=7}$ and $D_{N=6} < D_{N=5}$, the initial estimate $N = 6$ is validated as near-optimal for this path.